

CSCE 670 INFORMATION RETRIEVAL · MODULE 01 · LECTURE NOTES

The Information Need

The query is evidence, not a specification.

MODULE OUTCOMES

- Distinguish need, query, and document.
- Explain the vocabulary mismatch and its consequences.
- Explain the kinds of need and why they differ.
- Explain why relevance is not a property of a document.
- Set up a retrieval problem so it can be evaluated.

1 Three things, routinely conflated

	What it is	What the system actually sees
Information need	What the person wants to know, or to be able to do.	Nothing at all — it is never observed, at any point, by any means. See the note.
Query	Their attempt to express that need in a few words, typed into a box.	This, and only this — plus whatever context the system separately holds.
Document	Text that may or may not help them.	Its words, and whatever else you chose to index — links, metadata, click history.

The need is never observed, and that is the governing fact of the entire field — every metric, every relevance judgement, and every click is a proxy for something the system cannot see and never will. Which is what makes Module 05 genuinely hard rather than merely technical, and is why CSCE 676 Module 13 §3 identifies this course's proxy as a relevance label standing in for relevance to a person.

SO MATCHING THE QUERY WELL AND SATISFYING THE NEED ARE DIFFERENT OBJECTIVES

A system optimised to match query terms will return documents containing those terms — which is not the same thing as returning documents that answer the question, and the divergence between those two is where every retrieval failure lives.

And the person knows this, which changes their behaviour: they phrase their queries for the system rather than for a person — dropping function words, guessing at keywords, imitating what has worked before — so the query is already a translation, and frequently a poor one.

Which makes the query a noisy, truncated, strategically-chosen signal: two or three words standing in for a paragraph of context that the system never receives and the user never thought to supply.

So every technique in this course is an attempt to recover information the query omitted — from the collection's statistics, from the corpus's vocabulary, from other users' behaviour, or from a language model's knowledge — and that framing makes the course's sequence intelligible.

2 Vocabulary mismatch

SYNONYMY

the query says "car", the document says "automobile". A term match finds nothing at all.

POLYSEMY

the query says "jaguar". Three distinct needs behind one string, and term matching cannot separate them.

MORPHOLOGY

"running" and "ran" and "runs". Stemming helps, and over-stems ("university" -> "univers", "organisation" -> "organ").

SPECIFICITY MISMATCH

the query says "dog illness", the document says "canine parvovirus". Neither term appears in the other, and both are correct.

AND THE ASYMMETRY

the searcher does not know the collection's vocabulary -- which is frequently WHY they are searching. Expecting them to guess the right term is expecting them to already know the answer.

The asymmetry is the point of the slide, and it is worth stating plainly: **the person searching is the one least able to choose the right words**, because the words are in the material they have not read yet. **Which motivates query expansion (Module 09) and dense retrieval (Module 07) both**, and which explains why 'the user should have searched better' is never an acceptable diagnosis.

EVERY RETRIEVAL ADVANCE IS AN ATTACK ON VOCABULARY MISMATCH

Stemming and normalisation (Module 02 §2) attack the morphology row, crudely and very cheaply — and the over-stemming cost is real, which is why the aggressive stemmers fell out of favour.

Query expansion and relevance feedback (Module 09) **add terms the searcher did not supply**, drawn either from the collection's co-occurrence statistics or from the user's own interaction with the first results.

Latent methods and dense retrieval (Module 07) **represent meaning rather than words**, which addresses synonymy and specificity mismatch directly rather than by substitution — and is the same distributional argument as CSCE 638 Module 04's.

And learning to rank (Module 06) **uses features correlated with relevance that have nothing whatsoever to do with term overlap** — document quality, link structure, click history, recency — **which is a different attack on the same problem** and is why it was such a large advance.

3 Kinds of need

Intent	Example	What success means
Navigational	Find a specific site or document the user already knows exists.	One correct result, at rank 1. Anything else is a failure, which makes this the easiest intent to measure.
Informational	Learn about a topic.	Several useful results; and diversity among them helps, because redundancy is nearly worthless.
Transactional	Do something — buy, download, book, sign up.	A page where the action is actually possible — which is a property of the page rather than of its text.
Exploratory	Understand a space the user does not yet know the shape of.	Coverage and serendipity, measured over a session rather than over a single result list.

These need different metrics — reciprocal rank for navigational, graded cumulative gain for informational, and session-level measures for exploratory — which Module 05 §3 develops. Intent determining the metric is the practical consequence of this taxonomy, and it is why a single system-wide metric averages over queries that should be judged by different standards.

AND RELEVANCE IS A RELATION BETWEEN A DOCUMENT AND A NEED, NOT A PROPERTY OF THE DOCUMENT

The same document is relevant to one person and useless to another — depending on what they already know, what they are trying to accomplish, what reading level suits them, and what else they have already seen.

So it is contextual, graded, and dynamic: relevance depends on the session, declines sharply with redundancy, and is not binary — a partially useful document is the common case and is not captured by a yes/no label, which is why graded judgements exist.

And it is neither transitive nor stable. A document's relevance changes once the user has read a similar one, which means the relevance of a list is not the sum of its documents' individual relevances — a fact that nearly every metric assumes away (Module 05 §3).

Which is precisely why Module 05 is hard — every evaluation methodology in the field approximates all of this with a static, per-document, query-independent-of-session label, and the approximation is where the trouble starts.

4 Setting it up, and reading the course

- Write the queries from needs, not from documents. A query derived from a document guarantees that document is retrievable, so the evaluation measures whether you can find the thing you wrote the query from — which is circular, and is Project 2's first grading criterion for exactly that reason.

- **Write down your relevance criteria before judging anything** — because **judging first and articulating the criteria afterwards produces criteria that match what you happened to find**, which is CSCE 676 Module 11's problem in a judgement setting.
- **Decide how the results will be used**, since that determines the metric (§3) — one result, ten results, or a whole session, and each implies a different measure.
- **Set the latency budget**, because **it constrains the architecture** rather than merely the implementation (Module 04, and the two-stage design of Module 06 §4).
- **And identify the baseline you have to beat**, which is a properly tuned BM25 and **is considerably harder than it sounds** (Module 05 §4). **Queries from needs rather than documents is the one that invalidates evaluations**, and it is the easiest mistake to make when assembling a test set quickly.

HOW TO READ THIS COURSE

Modules 02 to 04 are the classical engine — the inverted index, the scoring functions and their derivations, and efficient query processing — **which is still the foundation of every production search system**, neural or otherwise.

Module 05 is the centre, because **every later claim in the course depends on it**, and because the evaluation methodology is where the field's assumptions are concentrated.

Modules 06 to 09 are the learned and semantic layer — learning to rank, dense retrieval, approximate vector search at scale, and query understanding — each attacking §2's mismatch from a different direction.

And Modules 10 to 13 are web scale, generative retrieval, consequences, and honest claims — **with Module 12 about what a ranking does to the things and people being ranked, which is not a side issue at scale** but a first-order property of a deployed ranking system.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Manning, Raghavan & Schütze, chapter 1 (free)	§1 and §2 , and the Boolean retrieval starting point that the rest of the course departs from.
Broder — A taxonomy of web search (free)	§3's taxonomy in the original — short, and the navigational/informational/transactional split still organises practice.
Saracevic — Relevance: A review of the literature (free)	§3's callout, developed at length — relevance as a relation, with the several decades of argument behind it.
Furnas et al. — The vocabulary problem (free)	§2 measured — how rarely two people choose the same word for the same thing, which is the quantitative basis for the whole module.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write five information needs as paragraphs**, then write the query you would type for each.
- 2 **Compare the two** and list what the query omitted.
- 3 **Ask two people to name one thing** and record how often they agree.
- 4 **Find a document you need** whose vocabulary you had to guess at.
- 5 **Classify twenty real queries** by intent, and note the ones that resist classification.
- 6 **Find a document relevant to one person and useless to another**, and say why.
- 7 **Judge the same document twice, a week apart**, and check whether you agreed with yourself.
- 8 **Write your relevance criteria** for Project 2's collection, before judging anything.
- 9 **Write thirty queries from needs** and verify none came from a document.
- 10 **Set your latency budget** and record it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Distinguish need, query, and document, and say what the system sees.
- 2 Why is the need never observed, and what follows?
- 3 Why are matching the query and satisfying the need different?
- 4 Name four forms of vocabulary mismatch and the asymmetry.
- 5 How does each retrieval advance attack the mismatch?
- 6 Name four query intents and what success means for each.
- 7 Why does relevance belong to a relation rather than a document?
- 8 Why is a list's relevance not the sum of its documents'?
- 9 Give five setup decisions, and the one that invalidates evaluations.
- 10 Why is Module 05 the centre of the course?